INDUSTRIAL QUALITY ASSISTANT

PHASE 1 RECIPE REPORT

API, Feedback, Schemas, Metrics, Governance and AI Security Validation

Server validation evidence and reproducible acceptance procedure

Document type	Phase 1 acceptance and validation recipe report
Project scope	IQA MLOps Phase 1
Validation target	Git integration between origin/main and origin/Alphabot42/security_threat_model
Execution platform	HP Z420 server reached through REMnux and Tailscale
Validation date	June 2026
Prepared for	Review and team reproducibility

Report purpose

This report records the acceptance validation performed for Phase 1. It is both an execution evidence report and a reproduction guide. It explains what was tested, why each control matters, how the server validation was executed, how outputs were interpreted, and which evidence must be kept for review.

Delivered evidence package

Deliverable	Purpose in the recipe package
IQA AI Security Threat Model Report	Security baseline used to define the Phase 1 controls and acceptance expectations.
tests/security/test_api_security_contracts.py	Dedicated security contract test suite covering the Phase 1 security controls.
docs/governance/iqa_phase1_governance_matrix.md	Governance traceability matrix linking controls, evidence and next steps.
HP Z420 validation outputs	Server side execution evidence proving that the integrated Phase 1 branch passes on the target environment.
Phase 1 Recipe Report	This document. It records the validation procedure, results and acceptance evidence.
Phase 1 Implementation Report	Main synthesis document to be updated last with the recipe report and server validation evidence.

Contents

1. Executive Summary

The Phase 1 validation was successfully executed on the HP Z420 integration server. The target validated by this report is not the local CubeAI development copy, but a server side integration branch created from the latest origin/main branch plus the Phase 1 branch origin/Alphabot42/security_threat_model. This validates the Phase 1 work in the context of the latest merged repository state.

Control area	Observed result	Acceptance status
Security contract tests	10 passed	Validated
API skeleton tests	21 passed	Validated
Full pytest suite	92 passed after temporary relocation of one ignored local checkpoint	Validated with documented environmental handling
Repository state after validation	Working tree clean, local ignored checkpoint restored	Validated
Pull request readiness	Code branch validated against current main through a local integration merge	Ready for PR preparation

2. Scope and Acceptance Criteria

The recipe covers the validation of Phase 1 controls related to the API contract, feedback lifecycle, schema constraints, security metrics exposure, governance traceability and server reproducibility. It does not certify the future Phase 2 integrations such as a live MLflow client, persistent PostgreSQL storage, a complete candidate dataset builder or Prometheus and Grafana dashboarding. These items remain outside the implemented Phase 1 boundary.

ID	Acceptance criterion	Validation method	Result
AC1	The HP Z420 server must be reachable through the approved private access path.	Tailscale ping and SSH connection	Tailscale peer visibility and SSH access confirmed.
AC2	Validation must be executed from the repository on the HP Z420 server.	Repository discovery and path check	Repository found under /opt/iqa/iqa-mlops.
AC3	The tested branch must include the current main branch and Phase 1 branch together.	Local validation branch and merge	Local validation branch created from origin/main and merged with origin/Alphabot42/security_threat_model.
AC4	The API and security test suites must pass.	Dedicated pytest commands	10 security tests and 21 API skeleton tests passed.
AC5	The full pytest suite must pass or any environmental blocker must be explained and handled safely.	Full pytest command and artifact diagnostic	Full suite passed after temporarily moving one ignored local .pt checkpoint outside the repo tree.
AC6	The server repository must be left clean after validation.	Git status and ignored artifact check	Working tree clean and ignored checkpoint restored.

3. Validation Approach

The validation deliberately separates the local development environment from the server acceptance environment. REMnux is used only as the SSH client and Tailscale endpoint. The tests are executed on the HP Z420 server, from the repository state available through GitHub. This avoids copying files from CubeAI and produces a cleaner acceptance proof.

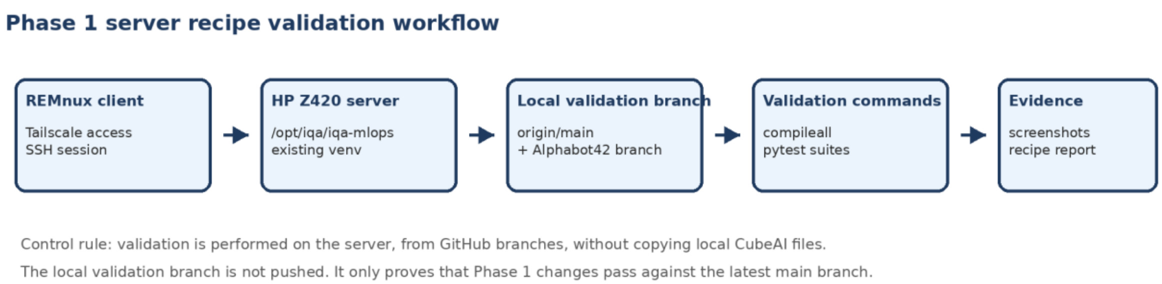


Figure 1. Server recipe validation workflow.

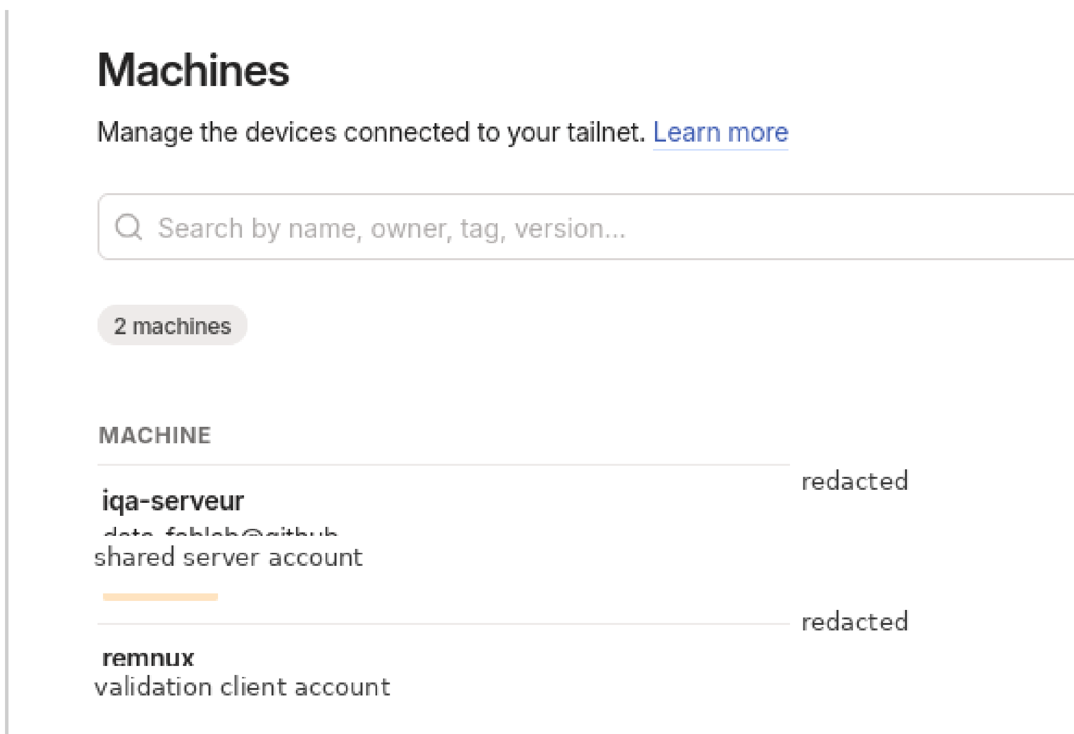


Figure 2. Tailscale machine visibility with sensitive information redacted.

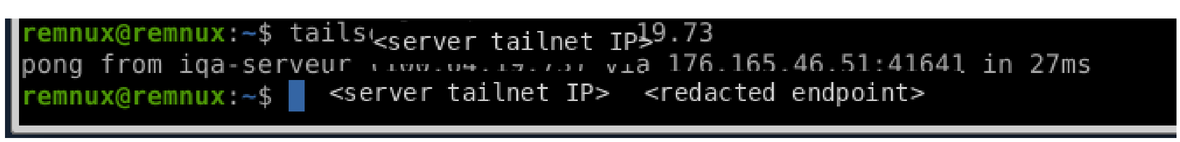
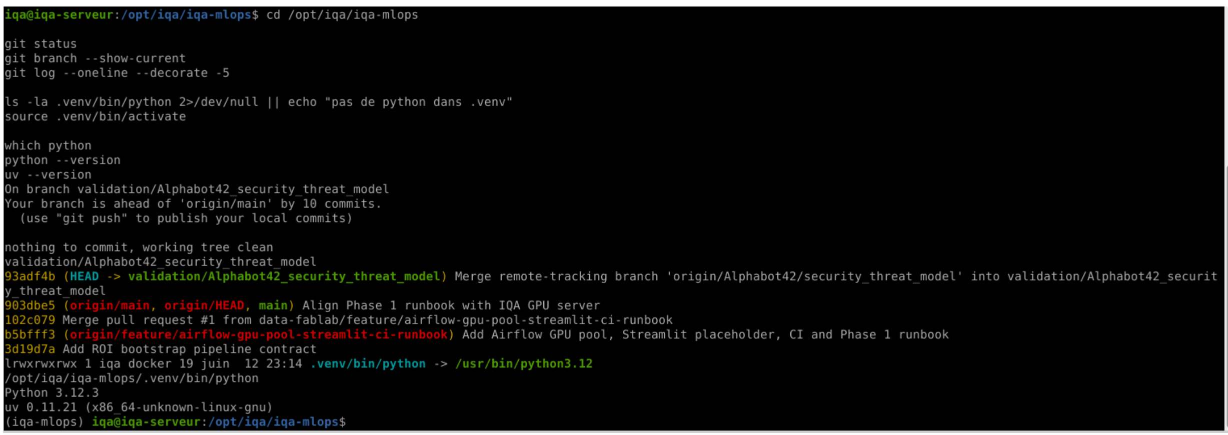


Figure 3. Successful Tailscale peer reachability test with addresses redacted.

4. Server Environment Baseline

Access path	REMnux VM connected to the HP Z420 server through Tailscale, then SSH using the project server account.
Repository path	/opt/iqa/iqa-mlops
Active Python runtime	/opt/iqa/iqa-mlops/.venv/bin/python
Python version	Python 3.12.3
uv version	uv 0.11.21
Server time state	UTC timezone, NTP synchronized. The timezone was not modified during this validation.
Disk state	Server had sufficient free space for validation. No large artifact was added to Git.

The server timezone was intentionally left unchanged. A future operational decision may align the server timezone to Europe/Paris, but this recipe records the environment as tested without modifying system configuration.



```
iqa@iqa-serveur:/opt/iqa/iqa-mlops$ cd /opt/iqa/iqa-mlops

git status
git branch --show-current
git log --oneline --decorate -5

ls -la .venv/bin/python 2>/dev/null || echo "pas de python dans .venv"
source .venv/bin/activate

which python
python --version
uv --version
On branch validation/Alphabot42_security_threat_model
Your branch is ahead of 'origin/main' by 10 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
validation/Alphabot42_security_threat_model
93adf4b (HEAD -> validation/Alphabot42_security_threat_model) Merge remote-tracking branch 'origin/Alphabot42/security_threat_model' into validation/Alphabot42_securit
y_threat_model
903dbe5 (origin/main, origin/HEAD, main) Align Phase 1 runbook with IQA GPU server
102c079 Merge pull request #1 from data-fablab/feature/airflow-gpu-pool-streamlit-ci-runbook
b5bfff3 (origin/feature/airflow-gpu-pool-streamlit-ci-runbook) Add Airflow GPU pool, Streamlit placeholder, CI and Phase 1 runbook
3d19d7a Add ROI bootstrap pipeline contract
lrwxrwxrwx 1 iqa docker 19 juin  12 23:14 .venv/bin/python -> /usr/bin/python3.12
/opt/iqa/iqa-mlops/.venv/bin/python
Python 3.12.3
uv 0.11.21 (x86_64-unknown-linux-gnu)
(iqa-mlops) iqa@iqa-serveur:/opt/iqa/iqa-mlops$
```

Figure 4. Server validation branch, virtual environment and runtime checks.

5. Git Integration Validation

The server repository was already present and clean on main. Because origin/main had been updated by another merged branch, the Phase 1 branch was not tested in isolation. A local validation branch was created from origin/main, then origin/Alphabot42/security_threat_model was merged into it. The branch was not pushed and exists only as a server side validation artifact.

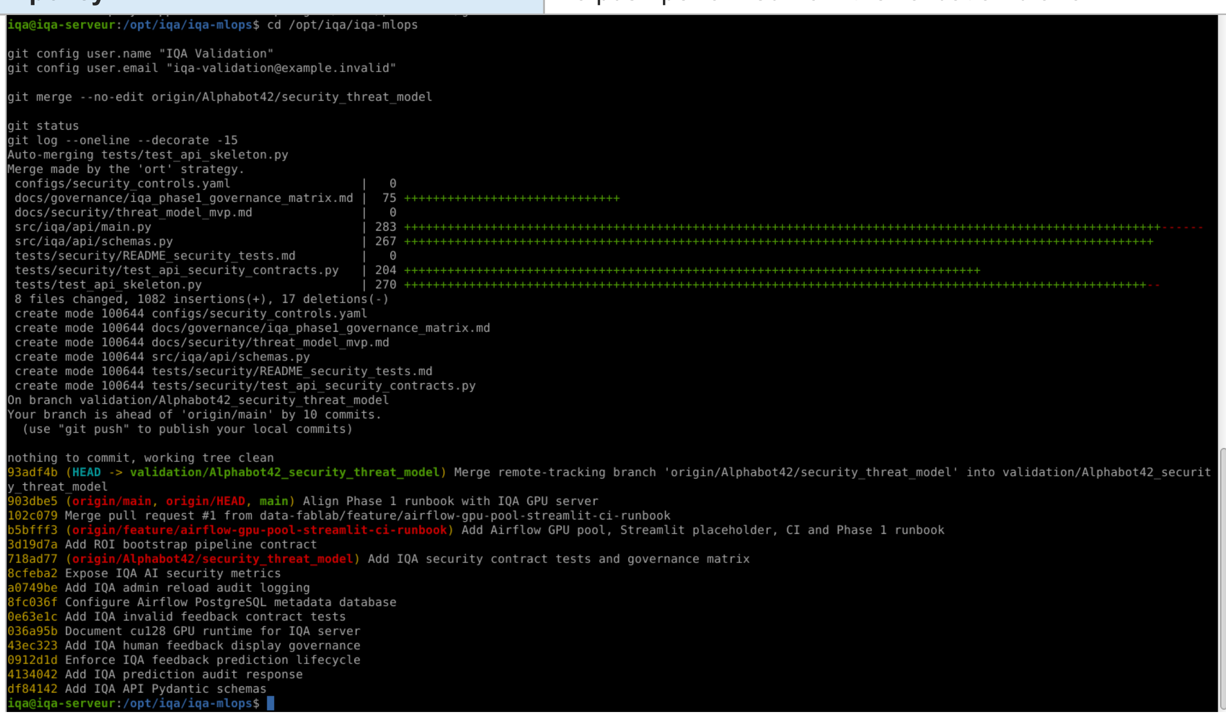
```
cd /opt/iqa/iqa-mlops

git switch main
git pull origin main

git switch -c validation/Alphabot42_security_threat_model origin/main
git config user.name "IQA Validation"
git config user.email "iqa-validation@example.invalid"
git merge --no-edit origin/Alphabot42/security_threat_model

git status
git log --oneline --decorate -15
```

origin/main	Latest server main branch including the recent integration updates.
origin/Alphabot42/security_threat_model	Phase 1 branch containing the API, feedback, schema, metrics and governance security work.
validation/Alphabot42_security_threat_model	Local server validation branch merging both lines of work.
Git identity	Repository local identity IQA Validation, used only to create the local merge commit.
Push policy	No push performed from the validation branch.



```
iqa@iqa-serveur:/opt/iqa/iqa-mlops$ cd /opt/iqa/iqa-mlops

git config user.name "IQA Validation"
git config user.email "iqa-validation@example.invalid"

git merge --no-edit origin/Alphabot42/security_threat_model

git status
git log --oneline --decorate -15
Auto-merging tests/test_api_skeleton.py
Merge made by the 'ort' strategy.
 configs/security_controls.yaml                |   0
 docs/governance/iqa_phase1_governance_matrix.md |  75 ++++++++++++++++++++++++++++++
 docs/security/threat_model_mvp.md             |   0
 src/iqa/api/main.py                           | 283 +++++++++++++++++++++++++++++++++
 src/iqa/api/schemas.py                        | 267 +++++++++++++++++++++++++++++++++
 tests/security/README_security_tests.md       |   0
 tests/security/test_api_security_contracts.py  | 204 +++++++++++++++++++++++++++++++++
 tests/test_api_skeleton.py                    | 270 +++++++++++++++++++++++++++++++++
 8 files changed, 1082 insertions(+), 17 deletions(-)
 create mode 100644 configs/security_controls.yaml
 create mode 100644 docs/governance/iqa_phase1_governance_matrix.md
 create mode 100644 docs/security/threat_model_mvp.md
 create mode 100644 src/iqa/api/schemas.py
 create mode 100644 tests/security/README_security_tests.md
 create mode 100644 tests/security/test_api_security_contracts.py
On branch validation/Alphabot42_security_threat_model
Your branch is ahead of 'origin/main' by 10 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
93adf4b (HEAD -> validation/Alphabot42_security_threat_model) Merge remote-tracking branch 'origin/Alphabot42/security_threat_model' into validation/Alphabot42_securit
y_threat_model
903dbe5 (origin/main, origin/HEAD, main) Align Phase 1 runbook with IQA GPU server
102c079 Merge pull request #1 from data-fablab/feature/airflow-gpu-pool-streamlit-ci-runbook
b5bfff3 (origin/feature/airflow-gpu-pool-streamlit-ci-runbook) Add Airflow GPU pool, Streamlit placeholder, CI and Phase 1 runbook
3d19d7a Add ROI bootstrap pipeline contract
718ad77 (origin/Alphabot42/security_threat_model) Add IQA security contract tests and governance matrix
8cfeba2 Expose IQA AI security metrics
a0749be Add IQA admin reload audit logging
8fc036f Configure Airflow PostgreSQL metadata database
0e63e1c Add IQA invalid feedback contract tests
036a95b Document cu128 GPU runtime for IQA server
43ec323 Add IQA human feedback display governance
0912d1d Enforce IQA feedback prediction lifecycle
4134042 Add IQA prediction audit response
df84142 Add IQA API Pydantic schemas
iqa@iqa-serveur:/opt/iqa/iqa-mlops$
```

Figure 5. Local validation merge between current main and the Phase 1 branch.

6. Validation Commands

The following commands represent the recipe executed on the server. They should be reused by colleagues to reproduce the acceptance validation after pulling the same branch combination.

```
source .venv/bin/activate

python -m compileall src/iqa/api tests/security

pytest -q tests/security/test_api_security_contracts.py
pytest -q tests/test_api_skeleton.py
pytest -q
```

7. Acceptance Test Matrix

ID	Control	Evidence source	Observed result	Status
T01	Security contract suite loads and executes	tests/security/test_api_security_contracts.py	10 passed	Pass
T02	API skeleton routes remain exposed	tests/test_api_skeleton.py	21 passed	Pass
T03	Prediction response exposes audit data	tests/test_api_skeleton.py	Covered in API skeleton test suite	Pass
T04	Feedback requires a valid prediction identifier	tests/test_api_skeleton.py and security contracts	Unknown prediction rejected	Pass
T05	Closed prediction replay is rejected	tests/test_api_skeleton.py and security contracts	Repeated feedback rejected	Pass
T06	human_sophie feedback is display only	tests/test_api_skeleton.py	Accepted without closing training feedback	Pass
T07	oracle_gt remains sovereign for training eligibility	tests/test_api_skeleton.py	Training source remains oracle_gt	Pass
T08	Invalid feedback source and status are blocked by contract validation	tests/security/test_api_security_contracts.py	Validation errors raised	Pass
T09	Admin reload requires configured admin token	tests/test_api_skeleton.py	Unauthorized and missing token cases logged and rejected	Pass
T10	AI security metrics expose Phase 1 counters	tests/test_api_skeleton.py	Metrics endpoint exposes expected counters	Pass
T11	Full repository test suite passes on HP Z420	pytest -q	92 passed after temporary relocation of ignored local checkpoint	Pass with documented environmental handling

8. Output Evidence and Interpretation

8.1 Phase 1 dedicated tests

The dedicated security contract suite and the API skeleton suite both passed on the HP Z420 server. This confirms that the Phase 1 controls are executable on the integration environment and not limited to CubeAI.

```
iqa@iqa-serveur:/opt/iqa/iqa-mlops$ cd /opt/iqa/iqa-mlops

git status
git branch --show-current
git log --oneline --decorate -5

ls -la .venv/bin/python 2>/dev/null || echo "pas de python dans .venv"
source .venv/bin/activate

which python
python --version
uv --version
On branch validation/Alphabot42_security_threat_model
Your branch is ahead of 'origin/main' by 10 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
validation/Alphabot42_security_threat_model
93adf4b (HEAD -> validation/Alphabot42_security_threat_model) Merge remote-tracking branch 'origin/Alphabot42/security_threat_model' into validation/Alphabot42_securit
y_threat_model
903dbe5 (origin/main, origin/HEAD, main) Align Phase 1 runbook with IQA GPU server
102c079 Merge pull request #1 from data-fablab/feature/airflow-gpu-pool-streamlit-ci-runbook
b5bfff3 (origin/feature/airflow-gpu-pool-streamlit-ci-runbook) Add Airflow GPU pool, Streamlit placeholder, CI and Phase 1 runbook
3d19d7a Add ROI bootstrap pipeline contract
lrwxrwxrwx 1 iqa docker 19 juin  12 23:14 .venv/bin/python -> /usr/bin/python3.12
/opt/iqa/iqa-mlops/.venv/bin/python
Python 3.12.3
uv 0.11.21 (x86_64-unknown-linux-gnu)
(iqa-mlops) iqa@iqa-serveur:/opt/iqa/iqa-mlops$
(iqa-mlops) iqa@iqa-serveur:/opt/iqa/iqa-mlops$
(iqa-mlops) iqa@iqa-serveur:/opt/iqa/iqa-mlops$
(iqa-mlops) iqa@iqa-serveur:/opt/iqa/iqa-mlops$
(iqa-mlops) iqa@iqa-serveur:/opt/iqa/iqa-mlops$
(iqa-mlops) iqa@iqa-serveur:/opt/iqa/iqa-mlops$ python -m compileall src/iqa/api tests/security

pytest -q tests/security/test_api_security_contracts.py
pytest -q tests/test_api_skeleton.py
pytest -q
Listing 'src/iqa/api'...
Compiling 'src/iqa/api/main.py'...
Compiling 'src/iqa/api/schemas.py'...
Listing 'tests/security'...
Compiling 'tests/security/test_api_security_contracts.py'...
..........                                                                [100%]
10 passed in 4.30s
.....................                                                     [100%]
21 passed in 4.20s
.....................................................................................F.......      [100%]
================================ FAILURES ================================
______________ test_no_heavy_model_or_sqlite_artifacts_in_repo_tree ______________

    def test_no_heavy_model_or_sqlite_artifacts_in_repo_tree() -> None:
        forbidden_suffixes = {".pt", ".pth", ".ckpt", ".onnx", ".sqlite"}
        offenders = [
            path
            for path in ROOT.rglob("*")
            if path.is_file()
            and path.suffix.lower() in forbidden_suffixes
            and ".venv" not in path.parts
        ]

>       assert offenders == []
E       AssertionError: assert [PosixPath('m...eckpoint.pt')] == []
E
E         Left contains one more item: PosixPath('models/roi_segmenter_v001_fixed/checkpoint.pt')
E         Use -v to get more diff

tests/test_repo_init_contract.py:138: AssertionError
========================== short test summary info ==========================
FAILED tests/test_repo_init_contract.py::test_no_heavy_model_or_sqlite_artifacts_in_repo_tree - AssertionError: assert [PosixPath('m...eckpoint.pt')] == []
1 failed, 91 passed in 13.33s
(iqa-mlops) iqa@iqa-serveur:/opt/iqa/iqa-mlops$
```

Figure 6. Dedicated Phase 1 tests passed and initial full suite blocker identified.

8.2 Full suite environmental blocker

The first full pytest execution did not fail because of the Phase 1 API or security controls. It failed because a local model checkpoint was present in the repository tree under models/roi_segmenter_v001_fixed/checkpoint.pt. The file was ignored by Git through .gitignore and was not tracked. The repository hygiene test scans for heavy model artifacts in the tree, therefore the local checkpoint had to be handled explicitly.

```
(iqa-mlops) iqa@iqa-serveur:/opt/iqa/iqa-mlops$ cd /opt/iqa/iqa-mlops

git status --short --ignored models/roi_segmenter_v001_fixed/checkpoint.pt
git ls-files models/roi_segmenter_v001_fixed/checkpoint.pt
git check-ignore -v models/roi_segmenter_v001_fixed/checkpoint.pt || echo "not ignored"
ls -lh models/roi_segmenter_v001_fixed/checkpoint.pt
!! models/roi_segmenter_v001_fixed/checkpoint.pt
.gitignore:11:models/**/*.pt    models/roi_segmenter_v001_fixed/checkpoint.pt
-rw-rw-r-- 1 iqa iqa 324M juin  13 20:39 models/roi_segmenter_v001_fixed/checkpoint.pt
(iqa-mlops) iqa@iqa-serveur:/opt/iqa/iqa-mlops$
```

Figure 7. Git diagnostic proving that the checkpoint is ignored and not tracked.

8.3 Safe handling of the ignored checkpoint

```
mkdir -p /tmp/iqa_validation_artifacts/roi_segmenter_v001_fixed
mv models/roi_segmenter_v001_fixed/checkpoint.pt
/tmp/iqa_validation_artifacts/roi_segmenter_v001_fixed/checkpoint.pt

pytest -q

mv /tmp/iqa_validation_artifacts/roi_segmenter_v001_fixed/checkpoint.pt
models/roi_segmenter_v001_fixed/checkpoint.pt
git status
git status --short --ignored models/roi_segmenter_v001_fixed/checkpoint.pt
```

The checkpoint was moved outside the repository only for the duration of the full test run. It was then restored to its original location to avoid changing the server runtime state. After the relocation, the full suite passed with 92 tests.

Figure 8. Full pytest suite passed after temporary relocation of the ignored checkpoint.

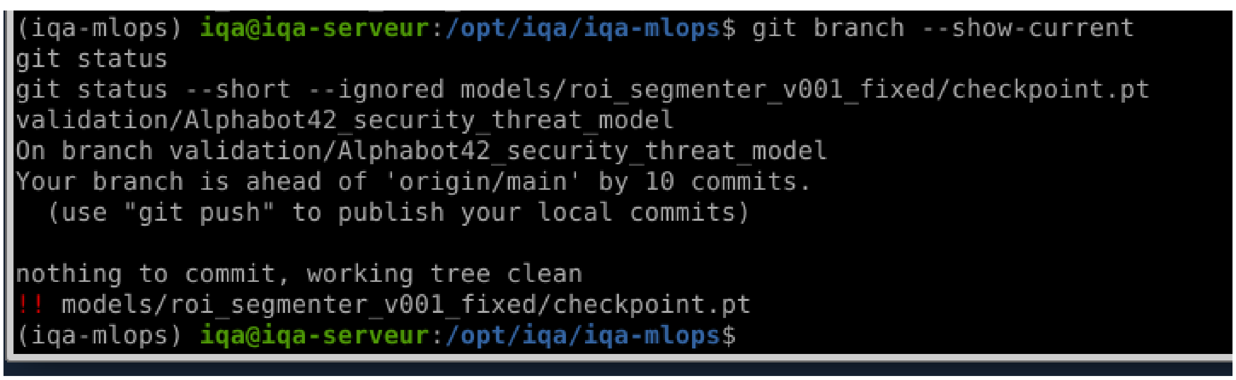


Figure 9. Final validation branch status after restoring the ignored checkpoint.

9. Findings and Acceptance Decision

ID	Finding	Decision
F01	The HP Z420 server can reach and validate the Phase 1 code through the approved access path.	Accepted
F02	The integrated branch composed of current main plus the Phase 1 branch merges without conflicts.	Accepted
F03	The dedicated Phase 1 security and API tests pass on the server.	Accepted
F04	The complete repository suite passes after removing a local ignored model artifact from the repository tree.	Accepted with note
F05	The local ignored checkpoint should be stored outside the Git working tree or managed through the project artifact strategy to avoid repeated hygiene failures.	Action recommended

Acceptance decision: Phase 1 is accepted for pull request preparation from a validation perspective. The test execution proves that the implemented controls pass on the HP Z420 integration server. The only blocker encountered was an ignored local model checkpoint already present on the server and unrelated to the Phase 1 branch content.

10. Reproduction Guide for Colleagues

This section provides the practical replay sequence for a colleague who needs to reproduce the validation in a real environment. The sequence assumes that Tailscale access and SSH credentials have already been provided by the infrastructure owner.

```
# 1. Connect to the server from REMnux
ssh iqa@<server_tailnet_address>

# 2. Enter the server repository
cd /opt/iqa/iqa-mlops

# 3. Verify baseline state
git status
git remote -v
git branch --show-current
git fetch origin

# 4. Create a local validation branch from current main
git switch main
git pull origin main
git switch -c validation/Alphabot42_security_threat_model origin/main

# 5. Merge the Phase 1 branch locally
git config user.name "IQA Validation"
git config user.email "iqa-validation@example.invalid"
git merge --no-edit origin/Alphabot42/security_threat_model

# 6. Run Phase 1 and full validation
source .venv/bin/activate
python -m compileall src/iqa/api tests/security
pytest -q tests/security/test_api_security_contracts.py
pytest -q tests/test_api_skeleton.py
pytest -q
```

If the full suite fails because an ignored local checkpoint is present under the repository tree, the file must not be committed. It can be moved outside the repository for the validation run, then restored if the server runtime expects it.

11. Risks, Limits and Follow Up Actions

ID	Risk or limit	Impact	Recommended action
R01	Ignored local model checkpoint inside the Git working tree	Can repeatedly break repository hygiene tests even though it is not tracked by Git.	Move runtime artifacts outside the repo tree or formalize DVC, MinIO or MLflow artifact storage.
R02	Server timezone currently set to UTC	May complicate human correlation with local CEST captures during troubleshooting.	Discuss Europe/Paris alignment with the infrastructure owner before changing system configuration.
R03	Local validation branch exists only on the server	It proves integration locally but is not a delivery branch.	Prepare a proper GitHub pull request from Alphabot42/security_threat_model to main.
R04	Phase 2 integrations are not covered by this recipe	Live MLflow, persistent storage and dashboards require separate validation.	Create dedicated Phase 2 recipe cases when those components are implemented.

12. Handover Checklist

- Server validation evidence captured and included in this report.
- Security contract tests passed on HP Z420.
- API skeleton tests passed on HP Z420.
- Full pytest suite passed after documented handling of the ignored checkpoint.
- Ignored checkpoint restored to avoid changing server runtime state.
- Repository working tree left clean.
- Main Phase 1 implementation report must now be updated with this recipe evidence.
- Pull request can be prepared after the main report update.

Appendix A. Command Log Summary

```
# Access validation
tailscale status
tailscale ping <server_tailnet_address>
ssh iqa@<server_tailnet_address>

# Server baseline
whoami
hostname
pwd
date
timedatectl
python3 --version
uv --version
find /opt /srv /mnt /home -maxdepth 6 -type d -name ".git" 2>/dev/null | sed 's#/.git##'

# Repository validation
cd /opt/iqa/iqa-mlops
git status
git remote -v
git branch -a
git log --oneline --decorate -10
git rev-list --left-right --count origin/main...origin/Alphabot42/security_threat_model

# Test validation
source .venv/bin/activate
python -m compileall src/iqa/api tests/security
pytest -q tests/security/test_api_security_contracts.py
pytest -q tests/test_api_skeleton.py
pytest -q
```

Appendix B. Evidence Register

Evidence ID	Evidence item	Description
E01	Tailscale machine visibility	Shows REMnux and the shared server in the tailnet, with sensitive data redacted.
E02	Tailscale ping	Shows successful peer reachability between REMnux and the server.
E03	Server environment	Shows validation branch, virtual environment and Python runtime.
E04	Local validation merge	Shows the merge of the Phase 1 branch into current main on a local validation branch.
E05	Dedicated tests	Shows 10 security tests and 21 API skeleton tests passing, with the first full suite blocker.
E06	Checkpoint diagnostic	Shows that checkpoint.pt is ignored by .gitignore and not tracked.
E07	Full suite pass	Shows 92 passed after temporary relocation of the ignored checkpoint.
E08	Final status	Shows working tree clean and checkpoint restored as ignored local artifact.

Appendix C. Terminology

Recipe report	Acceptance report that explains how validation was executed, which outputs were expected and which evidence proves acceptance.
Validation branch	Temporary local branch on the server used to test current main plus the Phase 1 branch without pushing any integration commit.
Ignored checkpoint	Local model artifact excluded from Git by .gitignore but still present in the server working tree.
Phase 1 boundary	Implemented controls for API contracts, feedback lifecycle, security metrics, schema validation, governance traceability and tests.